

SIGHETUL MARMATIEI, Romania

WMO: 150040

Lat: 47.9392N

Lon: 23.9044E

Elev: 276

StdP: 98.05

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.3	-18.1	0.8	-14.2	-15.4	1.0	-11.0	5.5	0.2	4.4	-0.5	1.1	140	0.457

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	31.4	20.7	29.8	20.0	28.2	19.2	21.9	29.0	20.9	27.7	20.1	26.5	1.5	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.8	24.6	18.7	14.0	23.6	17.8	13.2	22.9	65.1	29.2	61.7	27.8	58.7	26.5	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.0	4.0	3.3	-19.5	33.7	4.4	1.2	-22.6	34.6	-25.2	35.3	-27.7	36.0	-30.8	36.9		
			-19.7	23.8	4.3	0.9	-22.8	24.5	-25.3	25.0	-27.7	25.6	-30.8	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.9	-1.9	0.8	4.6	10.8	15.3	18.8	20.3	20.4	15.4	10.4	4.9	-1.3
	DBStd	8.93	4.66	4.45	4.10	4.18	3.34	3.21	2.93	3.07	3.87	4.16	4.53	5.20
	HDD10.0	1406	370	258	172	41	3	0	0	0	3	48	162	350
	HDD18.3	3297	628	491	427	226	103	32	15	14	104	246	402	608
	CDD10.0	1377	0	0	3	64	167	265	321	322	164	61	9	0
	CDD18.3	228	0	0	0	1	9	48	77	78	16	1	0	0
	CDH23.3	2301	0	0	0	23	140	477	722	756	175	9	0	0
	CDH26.7	710	0	0	0	2	21	141	232	271	44	0	0	0
Wind	WSAvg	1.1	0.9	1.0	1.4	1.4	1.2	1.2	1.2	1.1	1.0	1.0	1.0	0.8
Precipitation	PrecAvg	858	50	49	57	59	90	99	103	79	80	71	61	60
	PrecMax	1115	123	145	153	91	199	175	228	235	155	168	134	106
	PrecMin	652	22	19	9	21	25	35	21	19	19	8	5	9
	PrecStd	139	22	26	38	21	35	40	58	43	35	42	30	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	14.2	19.0	25.8	28.7	31.8	32.9	33.9	30.8	24.5	19.0	10.2
		MCWB	6.5	8.6	10.6	14.5	17.5	21.2	21.2	20.8	19.3	15.9	12.7	7.4
	2%	DB	6.9	10.6	15.3	23.3	26.8	30.1	30.9	31.7	27.5	22.1	16.1	8.0
		MCWB	5.1	6.8	9.0	13.8	16.9	20.1	20.9	20.6	17.6	15.1	11.3	6.1
	5%	DB	5.2	8.1	13.0	21.1	24.8	28.3	29.3	29.7	25.3	19.5	13.6	6.2
		MCWB	3.7	5.6	7.7	12.6	16.2	19.2	20.1	20.2	17.1	14.1	10.0	5.1
	10%	DB	3.6	6.2	10.7	18.8	22.9	26.1	27.5	27.6	22.6	16.7	11.3	4.9
		MCWB	2.4	4.1	6.7	11.4	15.4	18.3	19.6	19.3	16.2	12.9	8.9	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.2	11.7	15.7	19.2	23.1	22.7	23.3	20.1	16.8	13.4	8.1
		MCDB	8.5	14.0	16.9	23.4	26.1	30.0	29.9	30.5	28.5	22.2	17.4	10.2
	2%	WB	5.3	7.2	9.7	14.5	18.0	21.0	21.7	21.9	18.9	15.7	11.9	6.3
		MCDB	6.7	10.1	14.4	21.7	24.7	28.2	29.1	28.9	25.7	20.8	15.0	7.6
	5%	WB	3.8	5.9	8.4	13.3	17.1	20.1	20.9	20.8	17.9	14.6	10.6	5.3
		MCDB	5.0	7.9	12.1	19.8	23.2	26.5	27.6	27.4	23.5	18.6	13.4	6.3
	10%	WB	2.6	4.4	7.1	12.2	16.0	19.1	20.1	19.9	16.7	13.2	9.0	4.1
		MCDB	3.4	5.8	10.1	17.7	21.2	24.8	26.0	26.4	21.3	16.3	11.0	4.9
Mean Daily Temperature Range	5% DB	MDBR	6.4	7.9	9.8	11.8	12.1	11.9	11.8	12.7	11.1	10.0	7.6	6.1
		MCDBR	7.1	10.2	13.2	15.7	15.5	14.9	14.8	15.4	15.3	14.1	11.7	6.8
	5% WB	MCWBR	5.0	6.5	7.6	7.3	6.9	6.1	6.0	5.7	6.4	7.5	6.9	5.0
		MCWBR	6.3	9.3	11.6	14.3	13.7	13.5	13.3	13.8	13.3	12.5	10.2	6.4
Clear-Sky Solar Irradiance	taub		0.321	0.351	0.401	0.459	0.441	0.437	0.461	0.461	0.418	0.384	0.353	0.322
		taud	2.230	2.157	2.107	2.043	2.166	2.217	2.161	2.154	2.231	2.293	2.308	2.293
	Ebn at Noon		727	783	800	790	825	830	802	781	779	743	686	679
		Edh at Noon	80	107	132	156	143	137	143	138	117	94	75	67
All-Sky Solar Radiation	RadAvg		1.05	1.86	3.01	4.28	5.24	5.64	5.55	5.11	3.58	2.31	1.25	0.79
	RadStd		0.16	0.26	0.43	0.45	0.55	0.41	0.51	0.47	0.48	0.32	0.18	0.12

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (44 neighbors)	+0.70	N/A	N/A	+1.04	N/A	N/A	-150	-216	+105	+39	

Nomenclature: See separate page